DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Information Sheet 2023

The module Advanced Calculus (2AC) is one of the five modules offered within the second year course MAM2000W. Therefore, this information should be read in conjunction with the MAM2000W Course Information Sheet. The module Vula site created for this module is called 2AC-2023. Please make sure that you read the MAM2000W Course Information Sheet which contains the information regarding the administration of the course and the related half-courses. For instance, you should know the number of modules to be taken for the whole course and your half-course (if applied to you). You should also know the minimum requirement from your performance in the first semester modules, which will enable you to take the second semester modules. See the Vula website for the second-year Mathematics courses: https://vula.uct.ac.za/portal.

Advanced calculus also called multivariable calculus is the extension of the first year calculus to the study of vector valued functions of a single variable in connection with parametric curves, as well as the study of functions of several variables. The aim of this module is to provide tools and methods very useful in the description of important concepts in physical sciences, engineering, economics and computer graphics. The course content is attached to this module information sheet.

Prequisites: *You take this module because you have passed a first-year caclulus course which mainly involves among others the notion of vector and the geometry of space, the study of real-valued functions of a single variable involving the notions of differentiability and integration with their applications. Before you take this module, make sure that you had done a thorough revision of your first-year calculus.*

The plan through which we intend to deliver the module is described as follows.

Lectures:

- *There are two lecture groups which will take place in the venues M304 and M320 (in the Maths Building) on Tuesdays, Fridays and the following Wednesdays:* **Feb 22; Mar 8, 22; Apr 5, 19; May 17**
- *The class will be split alphabetically as follows*

Group	**Students**	**Venue**	**Day and Time**
Group I	A —> Mohabir	M304	Tuesday, Some Wednesday, Friday @12h00
Group II	Mokgabudi —> Z	M320	Tuesday, Some Wednesday, Friday @12h00

You will find on Vula the written notes for the week. There will also be lecture videos recorded in the venues during face-to-face lectures. However, notice that the videos will only be uploaded after that they have been fully processed. As an additional resource, you will also be using the textbook is Calculus: Concepts and Contexts, from 3rd Edition, Metric Version by James Stewart. The same book was prescribed for MAM1000W, MAM1005H and MAM1006H in 2022. The textbook provides among others a nice set of exercises that we recommend you solve.

- Two lecturers have been assigned to this module:

Dr François Ebobisse: Francois.EbobisseBille@uct.ac.za Office: M306
Ms Sané Erasmus: Sane.Erasmus@uct.ac.za Office: M405

Tutorials:
Tutorials are held *on Thursday and Friday afternoons in* 6^{th} *&* 7^{th} *periods* *You will be asked to choose a tutorial group according to your availability in those days. Each tutorial group will be under the responsablity of tutors who will assist you to solve the questions in the tutorial sheet during each session. You are expected to attend a one-period session on Thursday or Friday.* **Attendance to tutorials is compusolry!**

Assessment:
The assessment for this module will start from the assignment (AS) **online** *set up for the purpose of helping you start the class test 1 with a good performance.*

Below are the assessment details (date and time) while the syllabii for both tests and the assignment will be communicated later.

Test:	**Date**	**Time**
Assignment (AS):	*Monday 20th March*	*10h00-20h00 ;*
Class Test 1 (CT1):	*Wednesday 22nd March*	*18h00-19h30 .*
Class Test 2 (CT2):	*Tuesday 9th May*	*18h00-19h30 .*

Class record: *Your class record will calculated as follows:*

$$\left\{\begin{array}{c}\textbf{CTA:=AS + CT1,}\\ \textbf{CR:=0.5*CTA +0.5*CT2,}\\ \textbf{CTA is the combination of AS and CT1.}\end{array}\right.$$

Note:

1. *If you miss a class test without good reason you will be credited with a* **0** *(zero) for that test.*
2. *If you miss an assignment or a test for medical reasons, you must present a signed* original *medical certificate stating specifically that you were* medically unfit to write. *Medical certificates should be sent via e-mail to* *Francois.EbobisseBille@uct.ac.za or Sane.Erasmus@uct.ac.za* **no later than one week after the test**.

Examination and Final Mark:
The University is planning to hold final face-to-face examinations in June/July for the first semester courses/modules. According to that plan, the 2AC exam **(EX)** *will consist of a 2 hours paper.*

The **DP** *requirement for the 2AC module is a class record of at least 30% and 80% attendance to tutorials.*

Your class record **(CR)** *will count as either 20% or 40% of your final mark* **(FM)**, *whichever is to your advantage. Your examination mark will make up the balance. That is,*

$$\textbf{FM:=max(0.4*CR +0.6*EX, 0.2*CR+0.8*EX)}$$

Notice that, you will be required to score at least 45% as final mark in the 2AC moodule in order for you to continue with the module 2DE (Differential Equations) in the second semester!

Need Help? *Lecturers and tutors will be available via email or in-person. You can use the chatroom and the QA (more recommended) in the 2AC Vula site.*